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(12) **United States Design Patent**
Mirth

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(54) **ROBOT CONTROL UNIT**

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(52) **U.S. Cl.**

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(58) **Field of Classification Search**

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D14/454, 455, 496, 217, 218, 299;
D21/324–333, 566; D13/168; D15/199
CPC .. G05B 19/409; G05B 19/423; G05B 19/425;
G05B 19/427; G05B 2219/36159; G05B
2219/36162; G05B 2219/39443; G05B
2219/39001; G05G 9/047; B25J 13/06;
G06F 3/01; G06F 3/02; G06F 3/025;
G06F 3/033; G06F 3/041; G06F 3/0412;
G06F 3/0488; G06F 3/04886; G06F
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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

D279,489 S 7/1985 Mak
D307,415 S * 4/1990 Holmer D13/168
D338,452 S * 8/1993 Allgeier D14/374
D377,929 S * 2/1997 Sekino D14/336

D385,857 S 11/1997 Cohen et al.
D415,121 S * 10/1999 Uematsu D14/328
D432,137 S 10/2000 Holtzman et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 304732343 7/2018
CN 304904221 11/2018

(Continued)

OTHER PUBLICATIONS

3PE Teach Pendant For E-Series—Universal Robots. (online) 5 pgs.
Available—Last Modified Oct. 6, 2020. [Retrieved: Oct. 17, 2023]
<https://www.universal-robots.com/plus/products/universal-robots/3pe-teach-pendant-for-e-series/>.*

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(57)

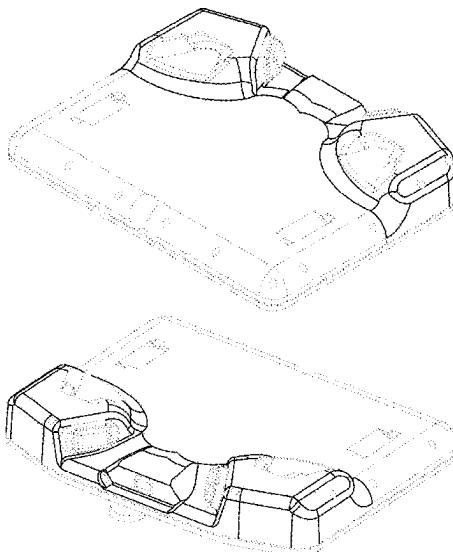
CLAIM

The ornamental design for a robot control unit as shown and described.

DESCRIPTION

FIG. 1 is a line drawing of a back, bottom perspective view of a robot control unit showing my new design;
FIG. 2 is a front, bottom perspective view thereof;
FIG. 3 is a bottom view thereof;
FIG. 4 is a front, down-facing view thereof;
FIG. 5 is a back, up-facing view thereof;
FIG. 6 is a right side view thereof, the left side being a mirror image; and,
FIG. 7 is a back, top, perspective view thereof.
The broken lines consisting of unevenly sized dashes define the boundary of the claimed surface, which extends to the boundary but does not include the boundary. All other broken or dotted lines in the figures depict portions of the robot control unit that form no part of the claimed design.

1 Claim, 7 Drawing Sheets



(56)

References Cited**U.S. PATENT DOCUMENTS**

6,134,102	A *	10/2000	Worn	B25J 13/06 361/679.09
D448,009	S	9/2001	Lavelle et al.	
D451,535	S	12/2001	Lee	
D461,802	S	8/2002	Tu	
D462,679	S	9/2002	Liu	
D464,345	S	10/2002	Liu	
6,494,527	B1	12/2002	Bischoff	
D480,720	S	10/2003	Sjoberg et al.	
D492,282	S *	6/2004	Lachello	D14/218
7,177,724	B2 *	2/2007	Cantello	B25J 13/06 700/61
D541,769	S	5/2007	Hoehn et al.	
D561,616	S	2/2008	Park et al.	
D577,724	S *	9/2008	Henne	D14/401
D581,818	S	12/2008	Gretton	
D592,214	S	5/2009	Bradford	
D600,727	S	9/2009	Apotheloz et al.	
D644,644	S *	9/2011	Wang	D14/415
D691,146	S	10/2013	Perez	
D703,664	S	4/2014	Spiro et al.	
D724,581	S	3/2015	Johnson	
D756,322	S	5/2016	Ostensen et al.	
D762,642	S	8/2016	Johnson et al.	
D770,052	S	10/2016	Abu-Tarif et al.	
D771,050	S	11/2016	Morgeneier et al.	
D773,405	S *	12/2016	Daubenmerkl	D13/168
D778,903	S	2/2017	Tsai	
D782,472	S *	3/2017	Tanaka	D14/341
D783,017	S	4/2017	Widiaman et al.	
D790,550	S	6/2017	Chen	
D803,221	S	11/2017	Guo	
D813,863	S	3/2018	Barea	
D825,497	S *	8/2018	Mizushi	D13/168
D827,646	S *	9/2018	Mitchell	D14/389
D835,100	S	12/2018	Zhan et al.	
D838,715	S	1/2019	Chen et al.	
D856,341	S	8/2019	Tur	
D864,137	S	10/2019	Chang et al.	
D868,708	S *	12/2019	Goos	D13/168
D868,709	S *	12/2019	Goos	D13/168
D868,710	S *	12/2019	Goos	D13/168
D869,405	S *	12/2019	Goos	D13/168
D871,401	S	12/2019	Jerregard et al.	
D871,589	S	12/2019	Oshima	
10,599,995	B2 *	3/2020	Tanaka	B25J 13/06
D881,894	S	4/2020	Zhang	
D884,701	S	5/2020	Huang et al.	
D884,702	S	5/2020	Huang et al.	
D887,403	S	6/2020	Matsushita	
D893,201	S	8/2020	Jiang	
D894,193	S	8/2020	Oberpriller et al.	
D894,405	S	8/2020	Li	
D897,535	S	9/2020	Rainville et al.	
D897,542	S	9/2020	Klock et al.	
D900,828	S	11/2020	Chen	
D902,202	S	11/2020	Fung et al.	
10,831,051	B2	11/2020	Jerregard et al.	
D907,212	S	1/2021	Pileski et al.	
D907,630	S	1/2021	Townsend, III et al.	
D910,615	S	2/2021	Wall et al.	
D924,228	S *	7/2021	Mirth	D14/341
D932,485	S *	10/2021	Mirth	D14/341
D932,486	S *	10/2021	Mirth	D14/341
D932,487	S	10/2021	Mirth	
D933,231	S *	10/2021	Marteau	D24/158

D941,369	S *	1/2022	Gehrmann	D15/138
D945,624	S *	3/2022	Marteau	D24/158
D962,441	S *	8/2022	Ooshima	D13/162
D1,029,840	S *	6/2024	Tur	D14/440
2017/0252004	A1	9/2017	Broad et al.	
2020/0171658	A1 *	6/2020	Kielsholm Thomsen	B25J 9/1664
2023/0418258	A1	12/2023	Mirth	

FOREIGN PATENT DOCUMENTS

CN	304958556	12/2018
CN	304981361	1/2019

OTHER PUBLICATIONS

3PE Teach Pendant For E-Series—Universal Robots. (online) 5 pgs. Available—Last Modified Oct. 6, 2020. [Retrieved: Oct. 17, 2023] <https://www.universal-robots.com/plus/products/universal-robots/3pe-teach-pendant-for-e-series/> lmg url :https://www.universal-robots.com/media/1814256/3pe-tp_gallery2.jpg.
Teach pendants for robots, announced Jul. 27, 2018 [online], [retrieved Oct. 8, 2020]. Available from Internet, URL: <<https://fabryka-robotow.pl/2018/07/teach-pendants-for-robots-the-changes-are-coming/>> (Year: 2018).
Robot companies, announced 2017 [online], [retrieved Oct. 8, 2020]. Available from Internet, URL: <<http://industrialrobot.info/robot-companies/>> (Year: 2017).
Kawasaki Robot, announced Jan. 25, 2017 [online], [retrieved Oct. 8, 2020]. Available from Internet, URL: <<https://www.youtube.com/watch?v=4enCu0LEStc>> (Year: 2017).
CS9 robot controller, announced 2020 [online], [retrieved Oct. 8, 2020]. Available from Internet, URL: <<https://www.staubli.com/en-gb/robotics/product-range/robot-controller/cs9-robot-controller/>> (Year: 2020).
iPad SP2 AED, announced 2020 [online], [retrieved Oct. 8, 2020]. Available from Internet, URL: <<http://defibzone.com/products/ipad-sp2-aed-standard-non-recharge.html>> (Year: 2020).
Robot teach pendants, announced Jan. 2010 [online], [retrieved Oct. 8, 2020]. Available from Internet, URL: <https://www.researchgate.net/figure/Robot-teach-pendants-a-the-MOTOMAN-NX100-teach-pendant-b-the-ABB-IRC5-teach-pendant_fig1_264237333> (Year: 2010).
Certificate of Registration for EP005272051-0001, 10 pages (May 18, 2018).
Certificate of Registration for EP005272051-0002, 10 pages (May 18, 2018).
Communication—Reply to official communication for EP005272051-0003, 2 pages (May 28, 2018).
Examination Report for EP005272051-0001/005272051-0004, 4 pages (May 24, 2018).
Receipt of an application for a registered Community design for EP005272051, 13 pages (May 18, 2018).
Replacement Figures as filed for CN 201830524587.5, 12 pages (Jun. 28, 2019).
Replacement Figures as filed for CN 201830524587.5, 12 pages (Feb. 20, 2019).
First Notification for Correction in Chinese Application No. 201830524587.5, 2 pages (Dec. 5, 2018), with English summary.
First Office Action in Chinese Application No. 201830524587.5, 1 page (Apr. 15, 2019), with English summary.

* cited by examiner

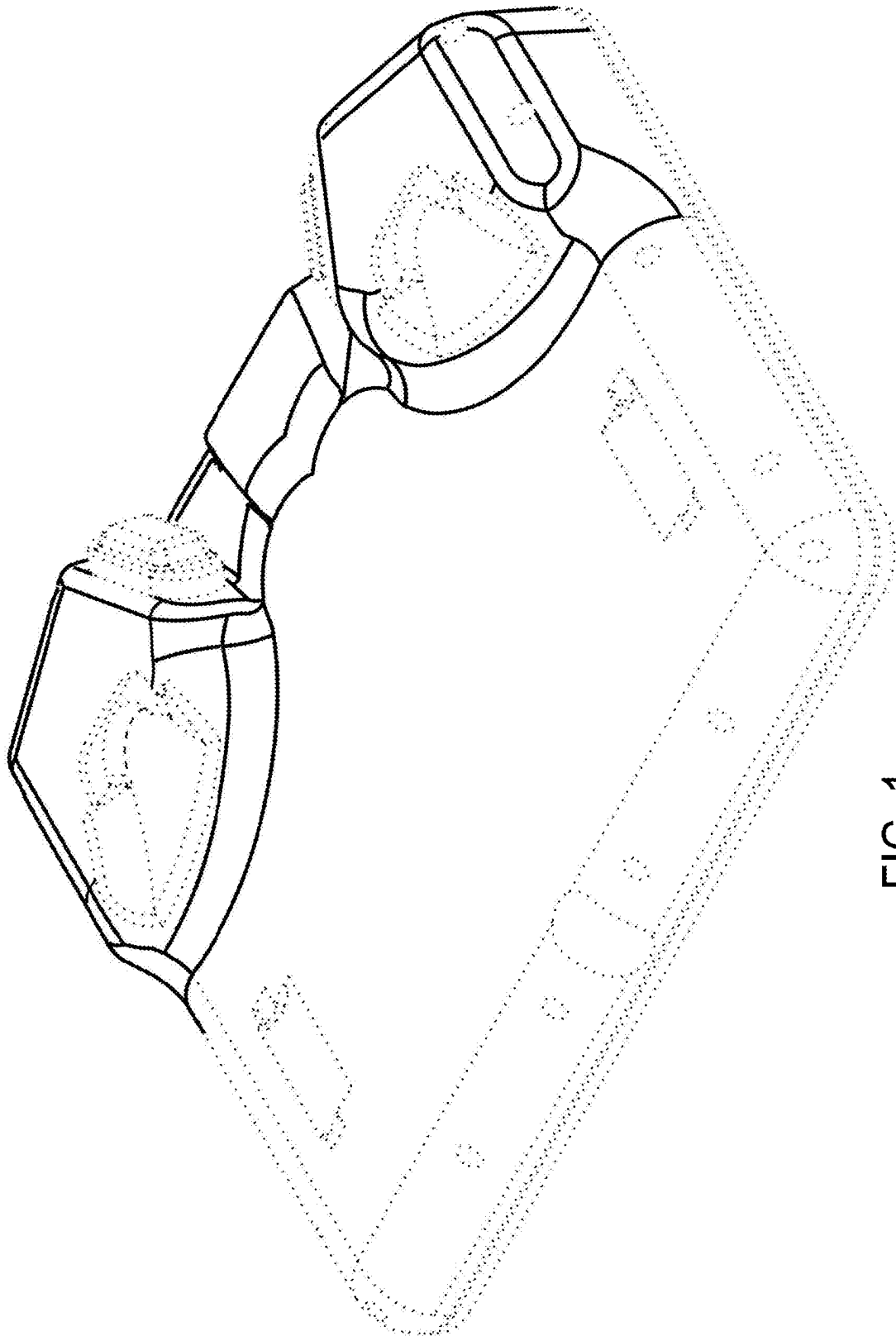


FIG. 1

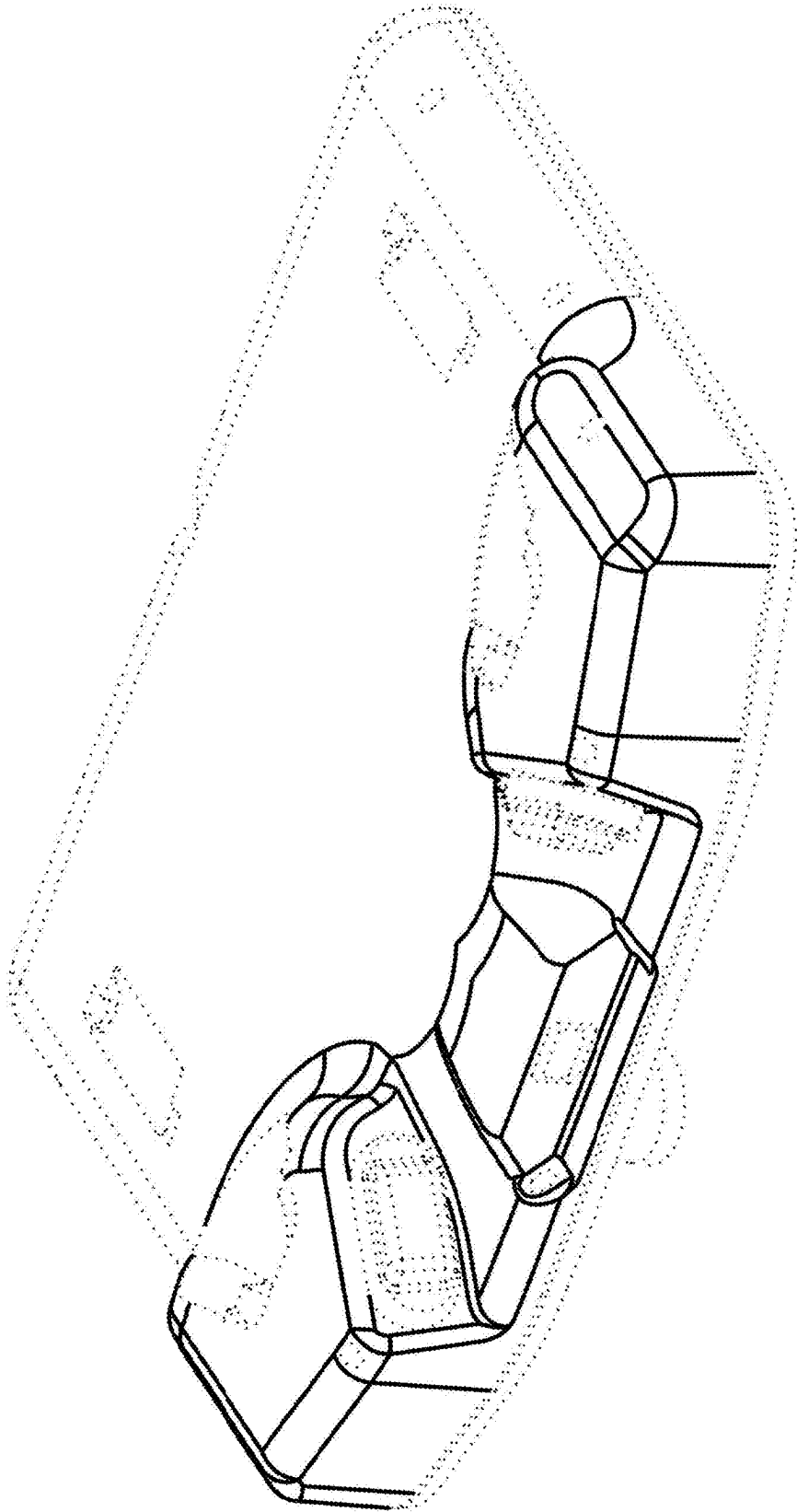


FIG. 2

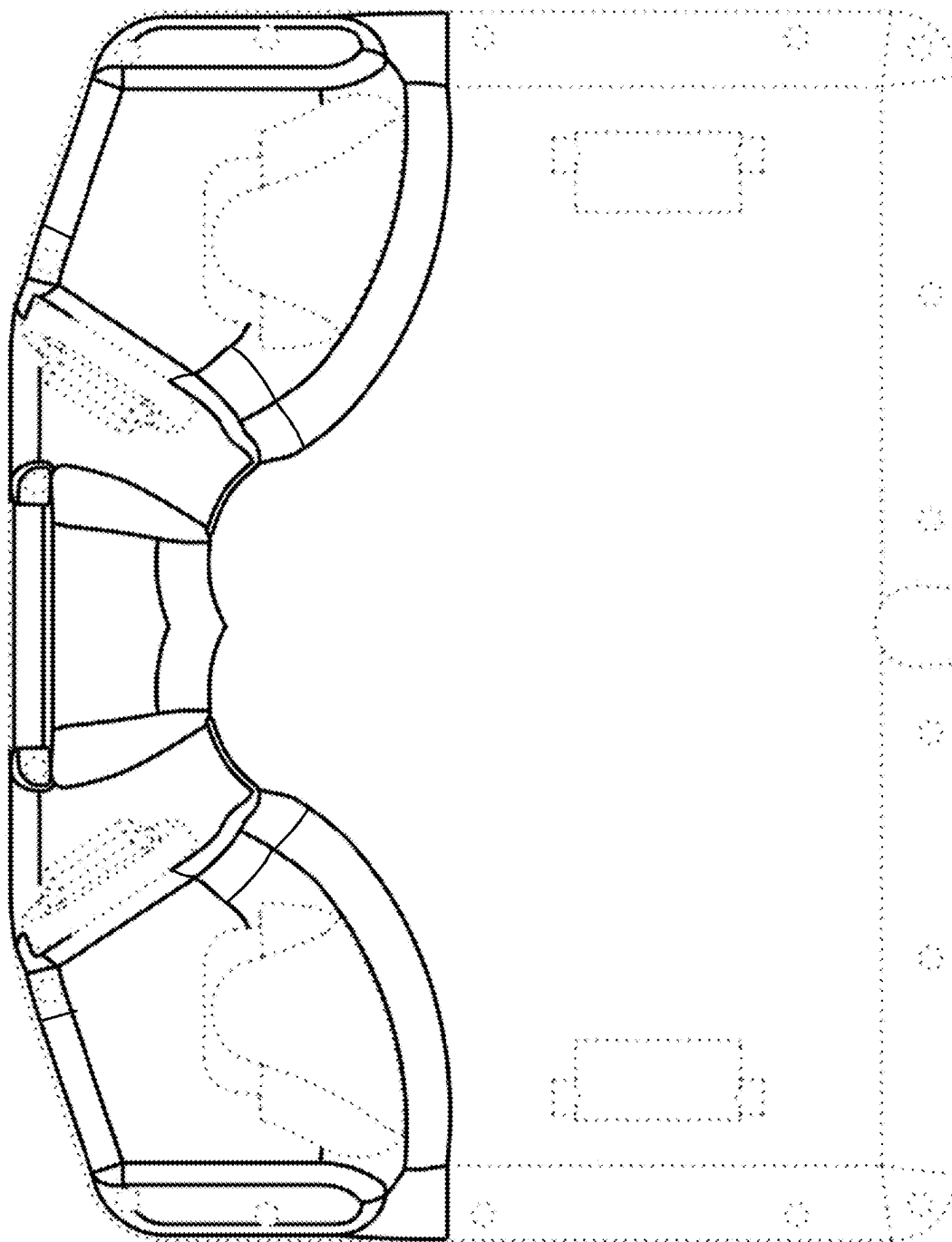


FIG. 3

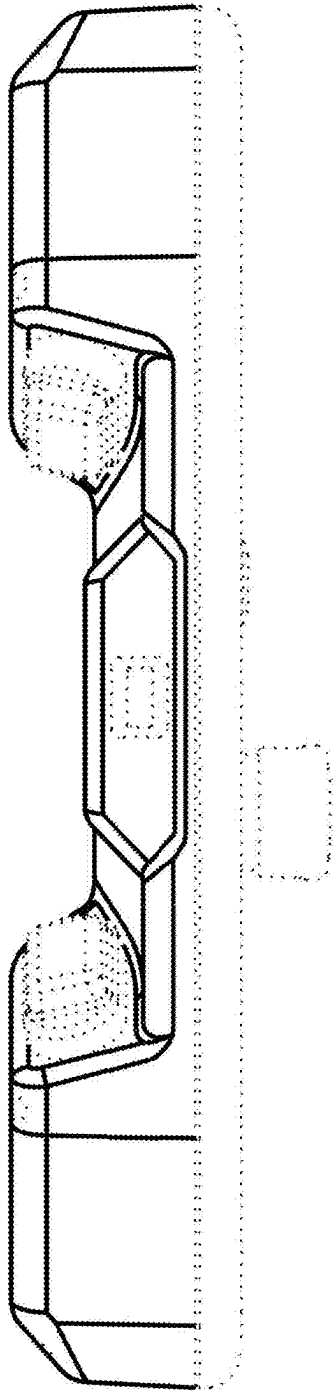


FIG. 4

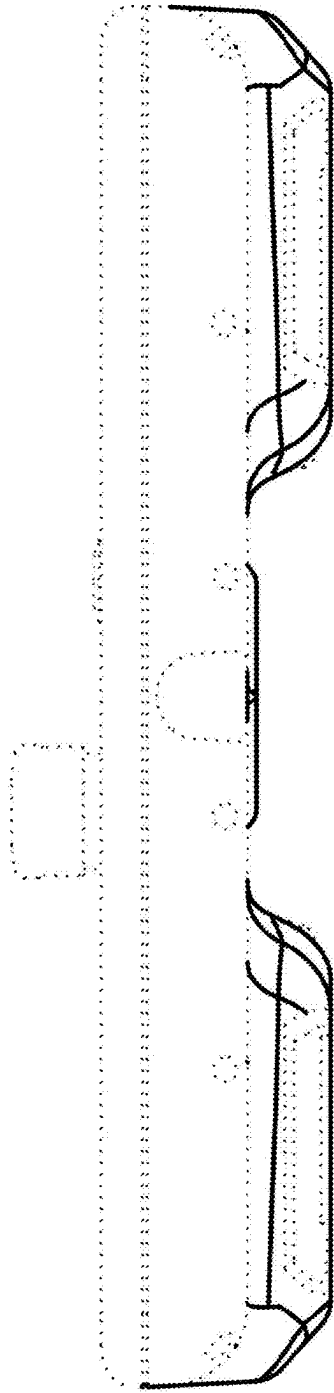


FIG. 5

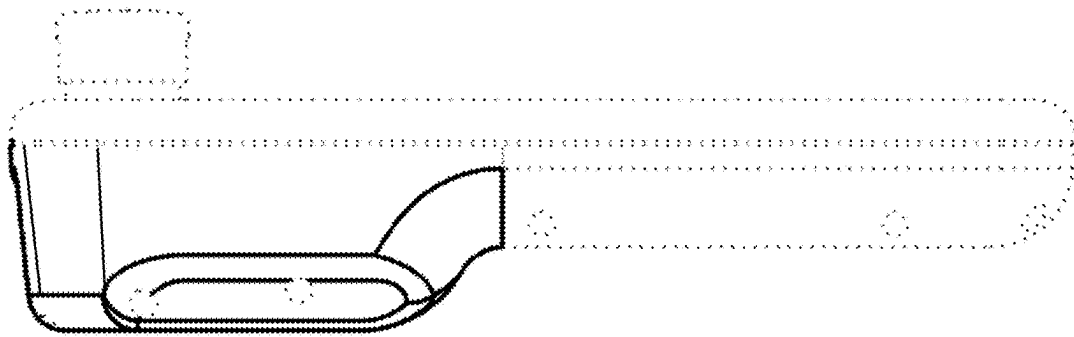


FIG. 6

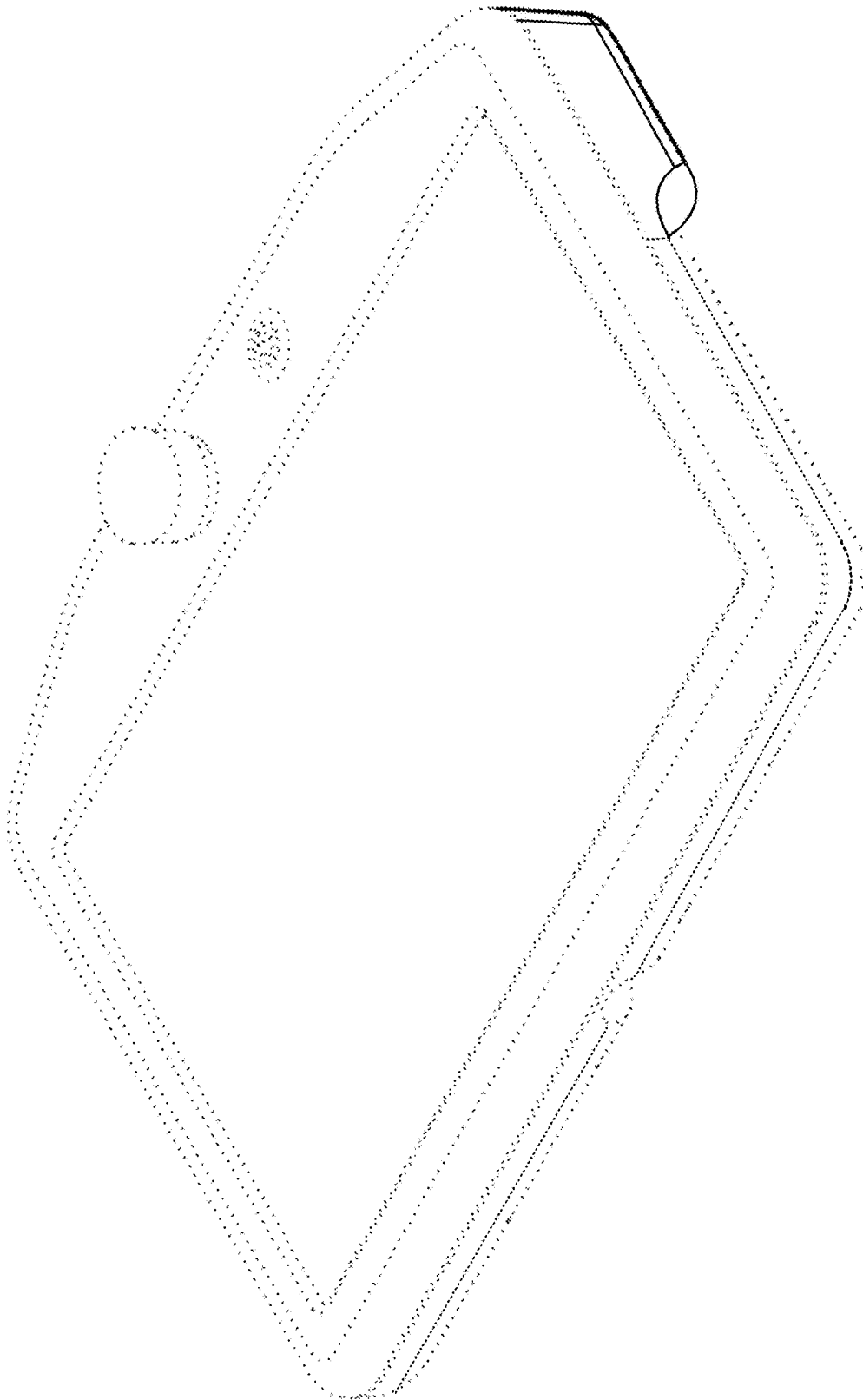


FIG. 7